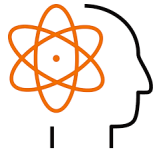


PSY301 SCIENTIFIC THINKING

Course Syllabus (CRN: 34464)



TR 4:00 - 5:20 PM

Global Scholars Hall 123,
University of Oregon

COURSE DESCRIPTION

Psychology is the scientific study of the mind and behavior, but what does it mean to study something *scientifically*? In this course, you will learn how psychologists measure constructs, validate their measures, design studies, select representative samples, follow ethical principles, collect data, make inferences about people from that data, and critically evaluate those inferences. You will learn to “follow the data” when people make claims about the human mind and behavior, wherever the data leads, and gain a set of skills to do it more wisely and rigorously, whether you are reading media accounts of research, reading peer-reviewed journal articles, or conducting scientific research yourself. By the end of the course, you will be able to read scientific research more fluently and critically, design your own studies more thoughtfully, and think more scientifically about questions related to the human mind and behavior. These are skills that are invaluable whether you wish to pursue a career that conducts or applies psychology research, another scientific field, or simply want to be a more informed consumer of research in everyday life.

OFFICE HOURS

Instructor

Sarah Dimakis, Ph.D. (“Sarah”)

sdimakis@uoregon.edu

W 2:00 – 3:30 PM & R 12:00 – 1:30 PM

Straub 437

Graduate Teaching Fellow

Lily Wang

xinywang@uoregon.edu

TBD

Graduate Teaching Fellow

Nathan Mejia Garduno

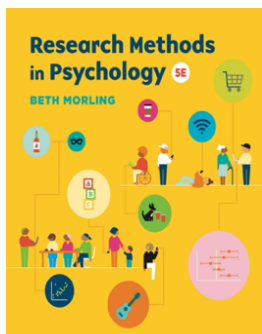
mejiagar@uoregon.edu

TBD

CONTACTING US

Please don’t hesitate to e-mail me or drop in on my office hours if you have any questions, concerns, or just want to chat. When you e-mail me, please add “scientific thinking” to the subject line (e.g., “scientific thinking exam question”) so that I know which class you’re from. I teach three classes per term (and 250 – 450 students per term), so this really helps me keep things organized and answer your question quickly. When you e-mail me, you can expect to receive a response within two days, and I typically respond same day (except weekends and holidays). If I don’t respond to your email within two business days, please resend it because I may have not received it.

COURSE MATERIALS



The course textbook is Beth Morling’s *Research Methods in Psychology*, 5th edition, though earlier editions are acceptable (and more affordable). All other readings or supplementary videos have been uploaded to Canvas. Some students like to read before lecture and others like to read after lecture. Either is completely fine. There is a lot of vocabulary in this class that you might be learning for the first time that will serve you well to retain. The repetition of reading and attending lecture is very important for both depth of understanding and retention of the concepts. I post all my lecture slides on Canvas prior to class and recommend downloading and annotating the slides during lecture.

COURSE OBJECTIVES

This course will help you develop skills that are valuable beyond university, such as introspection, critical thinking, and effective communication. You will also be working to improve skills that are necessary to be a consumer and/or producer of research, like reading about, critically evaluating, and communicating about scientific findings. Mainly, you will learn how to (1) think like a scientist who follows data and evidence, reducing your need to rely on the claims that others make about psychology research, (2) seek out psychology research that interests you and be able to identify the scientist(s)'s research questions, hypotheses, research design, evidence, and conclusions (3) critically evaluate the inferences that researchers (and the media) draw from scientific studies, and finally (4) communicate about psychological research, including the methodological and ethical issues currently plaguing the field.

ESTIMATED STUDENT WORKLOAD

Upon completion of this course, you will earn 4 credits toward your degree. A four-credit course from the University of Oregon is equivalent to 120 hours of work across the term, or 12 hours per week for 10 weeks. You will spend about 30 hours in class (3 hours a week), about 40 hours reading or watching supplemental material (4 hours a week), and the remaining 50 hours (about 5 hours per week) working on weekly assignments and studying for exams.

COURSE MODALITY

This is an in-person course with lecture integrated with practice problems and structured discussion. Unlike asynchronous online/ASYNC WEB courses, we will meet during scheduled class meeting times in *Global Scholars Hall 123*, which is on the East side of campus just off East 15th Ave and Moss St. Attendance is not graded but strongly encouraged. In class, you will be able to receive instantaneous feedback on whether you understand the concepts, ask questions when you're unsure, and connect with and build community with others in the class. For most students, you will learn more for less effort by coming to class. I do occasionally offer extra-credit opportunities that can only be earned if you are in class (four total, .5% each added to your final grade).

ASSIGNMENTS & GRADING

Your learning will be assessed through multiple choice and free-response exams (55% of your grade), research critique homework assignments (30% of your grade), and daily class worksheets (15% of your grade).

Assignment	Weight
Unit Exams (x3)	55%
Research Critique HW (x6)	30%
Class Worksheets (x18, drop 3)	15%
In-Class Extra Credit	2%

The following grid provides the letter grade associated with each percent. Grades within .1% of a grade category are rounded up, but not .11% or more. A student may be assigned an "A+" for a final grade if they demonstrate exemplary understanding of the course material across all assessments (> 97% on all three grade categories).

A 93-100%	B 83-86.99%	C 73-76.99%	D 63-66.99%
A- 90-92.99%	B- 80-82.99%	C- 70-72.99%	D- 60-62.99%
B+ 87-89.99%	C+ 77-79.99%	D+ 67-69.99%	F 59.99% or below

EXAMS

There will be three non-cumulative, in-person exams. Each exam has 30 multiple choice questions and five short answer questions that may ask you to interpret data, apply course concepts to examples, or generate examples based on a theory or model. You may bring a 3" by 5" notecard with handwritten or typed notes to use during the exam (front and back), food/ drink (try not to spill it on your exam though), and headphones if you want to listen to music or sounds (e.g., white or brown noise, café background noise, ocean sounds). A study guide will be posted **at least one week before each exam**, which will contain **all topics** that could be on the exam. There may be topics covered only in readings or videos (not in class), but those topics will be clearly specified on the study guide. Notecards will be provided in class the week before each exam, or you are welcome to use your own.

RESEARCH CRITIQUE HOMEWORK

Research articles are typically written by experts for other experts, which can make them very difficult to digest for anyone new to a field. Research critique homework assignments are designed to give you guided practice reading, dissecting, and critically evaluating empirical research articles, a skill that will be invaluable as you progress to advanced psychology courses. These guided exercises will, for instance, show you how to locate key information in an empirical journal article, apply recently learned course concepts to a real-world example, or ask you to critique different decisions that the researcher has made (e.g., measurement, sample composition). All homework assignments are due on **Canvas by 11:59pm** on the Sunday the week they are assigned.

CLASS WORKSHEETS

Active participation will help you retain the information you hear in lecture. Each class session is accompanied by a worksheet to guide you through applying the concepts we are learning that day. We will complete these worksheets during class in small groups and then review the answers together as a large class. Worksheets are due by **11:59pm** the day of class, and **three worksheets** will be dropped from your final grade to account for illness or unexpected life events. In case you need to miss more than 3 classes, you can complete the worksheet at home and upload it to Canvas. If you had to complete the assignment at home, you will be able to review the answer key when I post it the following day on Canvas. Class worksheets are graded based on completion: 1 (complete), .5 (partial), or 0 (incomplete).

LATE WORK & MAKE-UP EXAMS

To make it easy to remember, assignments in this course are always due on **Sundays at 11:59pm**. In the case that you fall ill or otherwise cannot complete your work on time, I have built flexibility into the course in the following ways:

- You can turn in your research homework assignments up to **seven days late for partial credit**.
- If an assignment is turned in **more than 24 hours** after the due date, the assignment will be marked off 5% (max score is now 95%). If 48 hours have passed, the assignment will be marked off 10% (max score is now 90%). Any homework after 72 hours will receive the max late penalty of 15% off (max score is now 85%) **until the assignment is locked seven days after the due date**.
- You have ONE late penalty waiver to turn in a research critique homework late, no questions asked. If you wish to use your waiver, comment on the assignment "I am using my late penalty waiver" when you submit it to Canvas. **A waiver cannot reopen a closed assignment**.
- If you need to miss a midterm exam, you must coordinate to retake it with a graduate teaching fellow. You must complete **Exam 1 by Thursday, April 30** and **Exam 2 by Tuesday, May 26** when exams are passed back in class and the answer key is posted on Canvas.

ACCESS AND ACCOMMODATIONS

The University of Oregon and I are dedicated to fostering inclusive, equitable, and accessible learning environments for all students. The Accessible Education Center (AEC) assists students with disabilities in reducing barriers in the educational experience. You may be eligible for accommodations for a variety of disabilities – apparent disabilities, such as a mobility or physical disability, or non-apparent disabilities, such as chronic illnesses or psychological disabilities. If you have or think you have a disability and experience academic barriers, please contact the Accessible Education Center (Location: 360 Oregon Hall; 541-346-1155; uoaec@uoregon.edu) to discuss appropriate accommodations or support. The details of your disability will be kept confidential with the AEC and you are not expected to share this information with others. However, I invite you to discuss any approved accommodations or access needs at any time with me.

ACCOMMODATIONS FOR RELIGIOUS OBSERVANCES

The University of Oregon respects the right of all students to observe their religious holidays, and will make reasonable accommodations, upon request, for these observances. If you need to be absent from a class period this term because of a religious obligation or observance and need an extension on a class worksheet, please fill out the Student Religious [Accommodation Request form](#) and send it to me within the first weeks of the course so we can make arrangements in advance.

CLASSROOM COMMUNITY EXPECTATIONS

It's imperative to a healthy learning environment that we treat each other with respect and commit to learning from each other. This requires us to (1) refrain from passing judgment on others (and ourselves) for making mistakes or not yet knowing something, (2) be cognizant of how much we are speaking and adjust accordingly to allow the class to hear from different perspectives. Students should be respectful of others' learning, including waiting to pack up until lecture is dismissed and sneaking in extremely quietly if you are late to class. Please sit in the back if you tend to get distracted or fall asleep in class, as to not distract others. Continued enrollment in the course indicates agreement that you will enthusiastically abide by these guidelines.

ARTIFICIAL INTELLIGENCE USE

You are permitted to use AI tools (e.g., ChatGPT, Grammarly) to assist with brainstorming ideas or refining your writing. However, keep in mind that AI can and often does provide inaccurate or overly simplistic responses, and you are responsible for the accuracy and quality of the work that you submit. If you choose to use these tools, then **they must not prevent you from achieving the learning goals of the course**. Relying on AI to think for you (such as copying a response directly from ChatGPT and submitting it as your own) will hinder your learning and therefore is considered a violation of the academic integrity policy for this course. We need to learn to strike a balance between using AI to make us more productive and losing or failing to gain important mental skills due to overreliance.

INTERESTED IN GRADUATE SCHOOL IN RESEARCH PSYCHOLOGY?

Please don't hesitate to come to office hours if you'd like to talk to me about graduate school in psychology. I am a recent (2024) graduate of UO's PhD program in Social and Personality Psychology and am always happy to share my experience (the highs and the lows). I can't speak too much to graduate school in mental health fields (more so to research psychology), but I am happy to give general advice and direct you to people who know more than me.

ACADEMIC INTEGRITY

Any form of academic dishonesty, including getting outside help on assignments, plagiarizing of any kind, improper use of AI (as outlined in the policy above), or lying to the instructor or a GTF will not be tolerated in this class. You are permitted (and encouraged) to work and study with other students on class assignments, but the work you submit *must be your own* and therefore should not be a direct copy of another student's work. If you have questions about what is considered academic dishonesty for this course, or if you are struggling with your coursework to the point that you feel you have no choice but to violate our academic integrity policy, please reach out to the instructor, the AEC, and/ or the dean of students. If I detect any academic dishonesty, I will follow the procedure to handle misconduct as outlined by the University, meaning that instances of suspected cheating or plagiarizing will be reported to the University.

YOUR WELLBEING

During your college experience, you may feel overwhelmed or stressed, experience anxiety or depression, struggle with relationships, or just need help navigating challenges in your life. If you're facing such challenges, you don't need to handle them on your own--there's help and support on campus. Getting help is a courageous thing to do—for yourself and those you care about. [University Health Services](#) helps students cope with difficult emotions and life stressors. If you need general resources on coping with stress or want to talk with another student who has been in the same place as you, visit the Duck Nest (located in the EMU on the ground floor) and get help from one of the specially trained Peer Wellness Advocates. University Counseling Services (UCS) has a team of dedicated staff members to support you with your concerns, many of whom can provide identity-based support. All clinical services are free and confidential. Find out more at counseling.uoregon.edu or by calling 541-346-3227 (anytime UCS is closed, the After-Hours Support and Crisis Line is available by calling this same number).

BASIC NEEDS

Being able to meet your basic needs is foundational to your success as a student. If you are having difficulty affording food, don't have a stable, safe place to live, or are struggling to meet another need, visit the [UO Basic Needs Resource Page](#) for information on how to get support. They have information food, housing, healthcare, childcare, transportation, technology, finances (including emergency funds), and legal support. If your need is urgent, please contact the Care and Advocacy Program by calling 541-346-3216, filling out the [Community Care and Support Form](#), or by [scheduling an appointment](#) with an advocate.

ACADEMIC DISRUPTION DUE TO CAMPUS EMERGENCY

In the event of a campus emergency that disrupts academic activities, course requirements, deadlines, and grading percentages are subject to change. Information about changes in this course will be communicated as soon as possible by email, and on Canvas. If we are not able to meet face-to-face, students should immediately log onto Canvas and read any announcements and/or access alternative assignments. Students are also expected to continue coursework as outlined in this syllabus or other instructions on Canvas.

INCLEMENT WEATHER

It is generally expected that class will meet unless the University is officially closed for inclement weather. If it becomes necessary to cancel class while the University remains open, this will be announced on Canvas and by email. Updates on inclement weather and closure are also communicated as described on the [Inclement Weather Webpage](#).

REPORTING OBLIGATIONS

I am a designated reporter. For information about my reporting obligations as an employee, please see [Employee Reporting Obligations](#) on the Office of Investigations and Civil Rights Compliance (OICRC) website. Students experiencing sex- or gender-based discrimination, harassment or violence should call the 24-7 hotline 541-346-SAFE [7244] or visit safe.uoregon.edu for help. Students experiencing all forms of prohibited discrimination or harassment may contact the Dean of Students Office at 541-346-3216 or the nonconfidential Title IX Coordinator/OICRC at 541-346-3123 to request information and resources. Students are not required to participate in an investigation to receive support, including requesting academic supportive measures. Additional resources are available at <https://investigations.uoregon.edu/how-get-support>. I am also a mandatory reporter of child abuse. Please find more information at [Mandatory Reporting of Child Abuse and Neglect](#).

PSY301 SCIENTIFIC THINKING

Spring 2026 Course Schedule

Week	Date	Topic	READING	DUE
1	T Mar. 31	Thinking like a scientist	Syllabus	
	R Apr. 2	Pseudoscience	Morling Ch. 1-2; Oreskes (2014)	
	S Apr. 5			
2	T Apr. 7	Variables and claims	Morling Ch. 3 (pp. 55 – 67)	
	R Apr. 9	A framework for evaluating scientific claims	Morling Ch. 3 (pp. 68 – 85)	
	S Apr. 12			HW 1
3	T Apr. 14	A history of ethical violations in scientific research	Morling Ch. 4 (pp. 89 – 93); Judo (2013)	
	R Apr. 16	Modern guidelines for ethical research	Morling Ch. 4 (pp. 94 – 115)	
	S Apr. 19			HW 2
4	T Apr. 21	EXAM 1		
	R Apr. 23	Collecting representative samples	Morling Ch. 7; Pew Research (2017)	
	S Apr. 26			
5	T Apr. 28	Developing reliable measures	Morling Ch. 5 (pp. 117 - 131)	
	R Apr. 30	Developing valid measures	Morling Ch. 5 (pp. 132 - 145)	
	S May 3			HW 3
6	T May 5	Self-report and observational methods	Morling Ch. 6	
	R May 7	Correlational studies	Morling Ch. 8; Vocativ (2015)	
	S May 10			HW 4
7	T May 12	Mediation, moderation, and third variables	Morling Ch. 9	
	R May 14	EXAM 2		
	S May 17			
8	T May 19	Simple experiments	Morling Ch. 10; Ted-Ed (2016)	
	R May 21	Threats to internal validity	Morling Ch. 11	
	S May 24			HW 5
9	T May 26	Factorial experiments	Morling Ch. 12	
	R May 28	Quasi-experiments	Morling Ch. 13	
	S May 31			HW 6
10	T Jun. 2	The replication crisis	Morling Ch. 14 (pp. 431 – 440); Veritasium (2016)	
	R Jun. 4	The credibility revolution	Morling Ch. 14 (pp. 441 – 458)	
	S Jun. 7			
Finals	Wednesday, June 10, 2026 at 12:30 p.m.			EXAM 3